



# UNIVERSITY OF GHANA

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BA/BSc. FIRST SEMESTER EXAMINATIONS: 2021/2022

DEPARTMENT OF STATISTICS & ACTUARIAL SCIENCE

STAT 339: METHOD OF LINEAR ALGEBRA (3 CREDITS)

INSTRUCTION: *Answer all questions in Section A and two questions from Section B.*

TIME ALLOWED: THREE (3) HOURS

Answer all questions  
(50 marks)

- A1. (a) Suppose  $U$  and  $V$  are subspace of a finite-dimensional vector space, show that

$$\dim U + \dim V = \dim(U \cap V) + \dim(U + V)$$

[10 marks]

- (b) A linear subspace  $U$  of  $\mathbb{R}^4$  is spanned by  $\{\vec{u}_1, \vec{u}_2, \vec{u}_3\}$  where

$$\vec{u}_1 = (1, 2, 3, 4), \vec{u}_2 = (-1, -1, 5, 0), \vec{u}_3 = (0, 0, 0, 1)$$

and another subspace  $V$  is spanned by  $\{\vec{v}_1, \vec{v}_2\}$ , where

$$\vec{v}_1 = (1, 0, 2, 3), \vec{v}_2 = (0, 1, -1, 2),$$

find the dimension of  $U$ ,  $V$ ,  $U \cap V$  and deduce  $\dim(U + V)$ .

[15 marks]

- A2. Let  $S = \{(u_1, u_2, u_3) : u_1, u_2, u_3 \in \mathbb{R}\}$ . For  $(u_1, u_2, u_3), (v_1, v_2, v_3) \in S$  and  $\lambda \in \mathbb{R}$ , define addition and multiplication by a scalar, respectively by

$$(u_1, u_2, u_3) + (v_1, v_2, v_3) = (v_1, u_2 + v_2, v_3)$$

and

$$\lambda(u_1, u_2, u_3) = (\lambda u_1, \lambda u_2, \lambda u_3).$$

- (a) Determine which of the axioms of a vector space are satisfied by  $S$  over  $\mathbb{R}$  and which ones are not. **[20 Marks]**

- (b) Is  $S$  a vector space over  $\mathbb{R}$ ? **[2 Marks]**

- (c) Show that

$$S = \{(x, y, z) : 2x + y - z = 0\}$$

is a subspace of  $\mathbb{R}^3$  **[3 Marks]**

- A3. (a) Show that the system of linear equations

$$\begin{aligned} 2x + y + z &= -6\beta \\ \gamma x + 3y + 2z &= 2\beta \\ 2x + y + (\gamma + 1)z &= 4 \end{aligned}$$

has a unique solution except when  $\gamma = 0$  and when  $\gamma = 6$ .

- (b) If  $\gamma = 0$ , prove that there is only one value of  $\beta$  for which a solution exists and find the general solution in this case.
- (c) Discuss the situation when  $\gamma = 6$ .
- (d) What are the conditions for inconsistency? **[25 Marks]**

- A4. (a) Show that every square matrix  $A$  can be written as the sum of a symmetric matrix and skew-symmetric matrix. Illustrate this by writing the matrix

$$A = \begin{bmatrix} 1 & 3 & 4 \\ 2 & 5 & -1 \\ 7 & -6 & 5 \end{bmatrix}$$

as the sum of a symmetric and a skew-symmetric matrix.

*Hint: Consider  $\frac{1}{2}[A + A^T]$  and  $\frac{1}{2}[A - A^T]$*  **[10 Marks]**

- (b) Prove that the set  $S = \{[1, -2, 2], [3, -4, -1], [1, -4, 9], [0, 2, -7]\}$  does not span  $\mathbb{R}^3$  **[6 Marks]**
- (c) If  $S = (v_1, v_2, \dots, v_r)$  is a set of vectors in a vector space  $V$  over a field  $F$ , define each of the following statements:
- i.  $S$  is linearly dependent over  $F$ .

- ii.  $S$  spans  $V$ .
- iii.  $S$  is a basis of  $V$ .

[9 Marks]

A5. A transformation  $T : V_3(\mathfrak{K}) \rightarrow V_2(\mathfrak{K})$  is defined by

$$T(x, y, z) = (x + y, y - z)$$

- (a) Prove that  $T$  is linear.
- (b) Find the representative matrix  $A$  of  $T$  relative to the pair of natural bases  $\{\vec{u}_1, \vec{u}_2, \vec{u}_3\}$  and  $\{\vec{v}_1, \vec{v}_2\}$  where

$$\vec{u}_1 = (1, 0, 0), \vec{u}_2 = (0, 1, 0), \vec{u}_3 = (0, 0, 1)$$

and

$$\vec{v}_1 = (1, 0), \vec{v}_2 = (0, 1),$$

- (c) Deduce from your result in (b) representative matrix  $A^*$  of  $T$  relative to the pair of bases  $\{\vec{u}_1^*, \vec{u}_2^*, \vec{u}_3^*\}$   $\{\vec{v}_1^*, \vec{v}_2^*\}$  where

$$\vec{u}_1^* = (1, 0, 0), \vec{u}_2^* = (0, 1, 0), \vec{u}_3^* = (0, 0, 1)$$

and

$$\vec{v}_1^* = (-1, 1), \vec{v}_2^* = (1, 2),$$

- (d) Compute
  - i.  $P_{V \rightarrow V^*}$
  - ii.  $R_{V^* \rightarrow V}$
  - iii.  $Q_{U \rightarrow U^*}$
  - iv.  $S_{U^* \rightarrow U}$
- (e) Verify that

$$A^* = Q^{-1}AP$$

[25 Marks]

A6. Let

$$A = \begin{bmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 0 \end{bmatrix}$$

- (a) Completely factorise  $|A - \lambda I_3|$  and find the eigenvalue of A.
- (b) For each eigenvalue in (a), find the corresponding eigenspace.
- (c) Find an orthogonal matrix  $P$  such as  $P'AP = D$ .
- (d) Hence, find the canonical form of the quadratic form  $Q : \Re^2 \rightarrow \Re$  given by

$$Q(x, y, z) = 4xy + 4xz + 4yz$$

[25 Marks]

A7. Using determinant or otherwise

- (a) Show that if  $a, b, c, d \in \mathbb{R}^+$ , then the system of equation

$$\begin{aligned}x + y + z + t &= a \\x - y - z + t &= b \\-x - y + z + t &= c \\-3x + y - 3z - 7t &= d\end{aligned}$$

has no solutions.

- (b) Find the value(s) of  $a$  such that the system of equations

$$\begin{aligned}3x + 2y + az &= 2a \\ax + y + 6z &= 15 \\x + 2y + 3z &= 8\end{aligned}$$

has infinitely many solutions.

[25 Marks]